Axiom at StanceEval-2026: A Two-Stage Hybrid Cascade for Arabic Stance Detection

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Abstract

Cross-target Arabic stance detection presents a unique challenge due to the rich morphological complexity, regional dialects, and heavy reliance on sarcasm within social media discourse. In this paper, we describe the system submitted by Team Axiom for Track 1 (Seen Targets) of the StanceEval-2026 shared task. Our primary contribution is a resource-efficient, Two-Stage Hybrid Cascade architecture. This system utilizes a fine-tuned MARBERTv2 encoder as a high-confidence gatekeeper, dynamically routing uncertain and complex tweets to a locally hosted, 4-bit quantized Large Language Model (Llama-3.1-8B-Instruct). To prevent label bias and ground the LLM's reasoning, the fallback stage incorporates a dynamically balanced, class-specific TF-IDF sub-word retriever. Our official submission achieved an F_{avg2} score of 0.6910 (Rank 18), demonstrating a measurable improvement over our standalone encoder baseline (0.6722).

1 Introduction

The proliferation of opinionated discourse on Arabic social media has accelerated the need for robust automated stance detection systems. Unlike standard sentiment analysis, which strictly assigns surface-level polarity (positive, negative, or neutral) (Mansour et al., 2025), stance detection requires predicting whether a writer is in favor of, against, or neutral toward a specific target—even when the opinion is expressed implicitly (Abu Farha and Magdy, 2020). The StanceEval-2026 shared task (Albalawi et al., 2026a), utilizing the Mawqif-XT dataset (Albalawi et al., 2026b), introduces the complex challenge of cross-target stance detection across highly diverse Arabic dialects.

Arabic dialectal stance detection is particularly difficult because the text is morphologically complex, informally written, and heavily reliant on implicit phrasing (Alkhathlan et al., 2025). Sarcasm and indirect phrasing cause measurable degradation in standalone sentiment systems, demonstrating why stance models cannot rely on surface polarity alone (Abu Farha and Magdy, 2020). Furthermore, domain transfer remains a core problem; even robust models struggle heavily on neutral stances and unseen topics (Alkhathlan et al., 2025).

Standard Transformer-based encoders, such as MARBERTv2, have historically set the benchmark for Arabic NLP due to their deep contextual understanding of primary dialects. However, these standalone models frequently suffer from softmax overconfidence, misclassifying sarcastic or highly colloquial text without any explicit reasoning mechanism. Conversely, modern Large Language Models (LLMs) possess the parametric depth to reason through nuanced dialectal idioms via In-Context Learning (ICL), but their heavy computational footprint makes processing large datasets sequentially prohibitive in resource-constrained environments.

In this paper, Team Axiom presents a multi-stage approach to Arabic stance detection that bridges the gap between encoder speed and generative reasoning. Our official StanceEval-2026 submission for Track 1 (Seen Targets) is a **Two-Stage Hybrid Cascade System**. We utilize a fine-tuned MARBERTv2 model as a primary gatekeeper, leveraging a strict confidence threshold to rapidly classify straightforward cases. Tweets falling below this threshold are isolated and routed to a 4-bit quantized Llama-3.1-8B-Instruct model operating on a single local consumer GPU, guided by a dynamically balanced sub-word retriever.

2 Related Work

Transformer Models for Arabic Dialects Standard Transformer-based encoders have historically set the benchmark for Arabic NLP. The literature clearly indicates that MARBERT-family models generally outperform ARBERT and AraBERT

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when dialectal and noisy user text is central to the task (AlShenaifi et al., 2024; Alghamdi et al., 2022; Hatem et al., 2025). This is largely attributed to pretraining data; MARBERTv2 is better aligned to social-media lexical variation, whereas ARBERT is more MSA-oriented (Hatem et al., 2025). Specifically, for stance detection, fine-tuned MARBERT has consistently beaten AraBERT baselines (AlShenaifi et al., 2024).

LLM Classification Limits and Retrieval While Large Language Models (LLMs) are powerful, their use in large-scale text classification is constrained by context limits, computational cost, and inconsistent performance on irony and contrast (Sun et al., 2023; Wang et al., 2024). To address these limitations, recent work has shifted toward retrieval augmentation, which reduces the need to encode all knowledge in parameters alone (Izacard et al., 2022). Dynamic demonstration retrieval with kNN search has proven highly effective at overcoming context-window limits by selecting task-relevant examples for prompting (Sun et al., 2023).

Hybrid and Cascaded Architectures Current architectural paradigms increasingly combine transformers, retrieval, and cascaded validation rather than choosing one model family alone. Hierarchical pipelines, such as two-stage architectures and cascaded embedding-plus-LLM validation designs, have shown high efficacy in Arabic classification (Almalki et al., 2026; Ilman et al., 2025). Our work directly builds upon these findings by utilizing a fine-tuned encoder for core classification, with a retrieval-augmented LLM fallback utilized as an add-on mechanism for ambiguous edge cases.

3 System Overview

Our official submission architecture is a resource-efficient Two-Stage Hybrid Cascade. This section details both the foundational encoder and the dynamic LLM fallback pipeline. The complete architecture is illustrated in Figure 1.

3.1 Stage 1: Baseline Encoder

For the primary classification layer, we utilized MARBERTv2 (Abdul-Mageed et al., 2021). We attached a standard sequence classification head to output probability distributions across the three stance classes (*Favor, Against, None*). While computationally efficient, early validation revealed that the standalone model suffered from softmax overconfidence, frequently assigning high probabilities

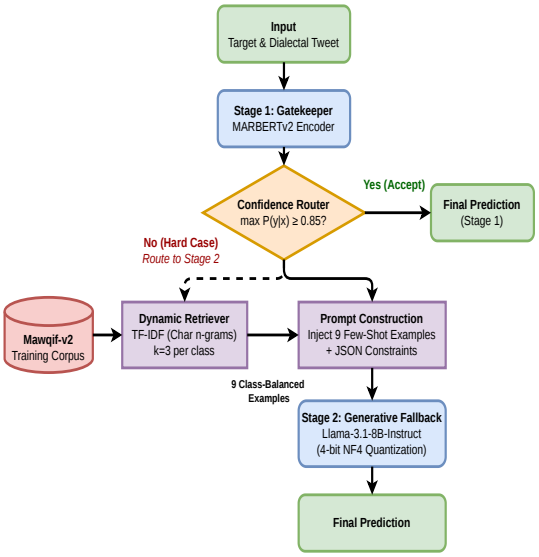


Figure 1: Architecture of the Two-Stage Hybrid Cascade. The MARBERTv2 gatekeeper rapidly classifies high-confidence tweets, while uncertain cases ($\tau < 0.85$) are routed to a retrieval-augmented LLM.

(> 0.85) to texts containing subtle sarcasm or complex Gulf-region idioms without executing nuanced dialectal reasoning.

3.2 Threshold-Based Dynamic Routing

To mitigate encoder overconfidence without incurring the massive latency of processing every tweet through an LLM, we implemented a threshold-based routing system. Let $P(y|x)$ be the probability distribution generated by MARBERTv2. We defined a strict confidence threshold $\tau = 0.85$. This value was empirically selected during validation to optimize the trade-off between maintaining baseline encoder accuracy and minimizing the computational latency of invoking the generative fallback. If the maximum predicted probability $\max P(y|x) \geq \tau$, the system accepts the encoder's prediction. If $\max P(y|x) < \tau$, the tweet is flagged as a "hard case" and routed to Stage 2.

3.3 Dynamic Balanced Class Retriever

To prevent label bias during LLM In-Context Learning (ICL), we built a dynamic retrieval mechanism using Term Frequency-Inverse Document Frequency (TF-IDF) operating on character n-grams. For every hard-case tweet routed to the LLM, the retriever filters the training corpus by the current target topic and extracts exactly $k = 3$ examples per class (3 *Favor*, 3 *Against*, 3 *None*). This ex-

poses the LLM to a perfectly balanced decision boundary, isolating the linguistic nuances of the target topic.

3.4 Stage 2: Quantized LLM Fallback

To execute the reasoning phase strictly on local consumer hardware without API dependency, we utilized 4-bit NormalFloat (NF4) quantization via BitsAndBytes. We deployed the open-weights `Meta-Llama-3.1-8B-Instruct` model, parsing the dynamically retrieved few-shot examples through a rigorous system prompt demanding JSON-formatted outputs(see Appendix A). The LLM evaluates the hard cases directly, overriding the uncertain encoder predictions with contextually reasoned stance labels.

4 Experimental Setup

4.1 Data and Preprocessing

We utilized the official Mawqif-XT dataset splits. For Track 1 (Seen Targets), the training set consists of 3,502 tweets and the validation set contains 619 tweets distributed across three thematic targets: COVID-19 Vaccine, Digital Transformation, and Women Empowerment. The input formulation pairs a target topic with a dialectal tweet, and the system outputs a single stance label (*Favor, Against, or None*). Minimal preprocessing was applied to preserve raw dialectal features; URLs and user mentions were normalized, while hashtags, emojis, and repeated characters were retained as pragmatic signals for stance intensity.

4.2 Hyperparameters and Hardware

The MARBERTv2 baseline was fine-tuned for 3 epochs with a learning rate of 2×10^{-5} and a batch size of 32. For the LLM generation, temperature was strictly set to 0.0 to ensure deterministic outputs. The entire Hybrid Cascade pipeline was executed natively on a single NVIDIA T4 GPU (15GB VRAM) available via Google Colab.

5 Results and Ablation

5.1 Ablation: Standalone Encoder Baseline

To measure the efficacy of our cascade architecture, we first evaluated our standalone MARBERTv2 model. On the official test set, this purely encoder-based approach achieved an overall accuracy of 0.6335 and an F_{avg2} score of 0.6722. Notably, the baseline struggled with the ambiguous "None" class, yielding a lower F_{avg3} score of 0.4633.

5.2 Official Submission Performance

By delegating the uncertain cases ($\tau < 0.85$) to the 4-bit Llama-3.1-8B fallback mechanism, our official submission (Team Axiom) secured Rank 18 on the final leaderboard. The generative fallback mechanism improved overall accuracy to 0.6534 and pushed our primary F_{avg2} score to 0.6910.

Most significantly, as detailed in Table 1, the Hybrid Cascade resulted in a substantial $\sim$4% absolute improvement in the F_{avg3} metric (0.5032). This confirms that incorporating dynamically retrieved, balanced few-shot prompts enables generative models to successfully resolve the complex, neutral, and dialectal nuances that traditional encoders misclassify.

Model	Accuracy	F_{avg2}	F_{avg3}
Baseline: MARBERTv2	0.6335	0.6722	0.4633
Final	**0.6534**	**0.6910**	**0.5032**

Table 1: Official StanceEval-2026 test set results. The Hybrid Cascade demonstrates consistent performance uplift across all metrics, with a notable improvement in the stringent F_{avg3} score.

6 Error Analysis

Despite architectural improvements, cross-target stance detection remains challenging. A manual review of the validation set revealed recurring failure modes within the generative fallback stage:

Sarcastic Slang and Monikers: Generative models occasionally failed to recognize heavy sarcastic slang. For example, regarding the Covid Vaccine target, the phrase عزيزي ابو جرعتين لا تتأخر الجرعة التنشيطية لما يوصلك مسج (*Dear Mr. Two-Doses, when you get the booster text do not delay*) was interpreted by the LLM as a **Favor** stance due to the literal instruction to "not delay." However, the use of the sarcastic moniker "Abu Jare'tein" and the subsequent mocking of government privileges in the full tweet indicates a strong **Against** stance.

Target-Agnostic Negative Sentiment: In cases where heavy negative sentiment was present, the LLM sometimes struggled to determine the actual object of the hostility. For instance, the tweet لقاح كورونا حول تطعيم كلمات المرجفين والجبناء (*The words of fearmongers and cowards regarding the corona vaccine*) was misclassified as **Against**.

The generative system erroneously associated the negative phrasing (cowards/fearmongers) with the vaccine itself, rather than realizing the author was attacking the vaccine's critics, which actually denotes a **Favor** stance.

7 Conclusion

In this paper, we detailed our methodology for the StanceEval-2026 shared task. By architecting a zero-cost, locally hosted Two-Stage Hybrid Cascade, we successfully bridged the gap between encoder efficiency and LLM reasoning. Combining threshold-based routing, class-balanced dynamic retrieval, and 4-bit quantization allowed our Llama-3.1-8B fallback to measurably improve our baseline MARBERTv2 score. Future work will explore applying Low-Rank Adaptation (LoRA) on generated Chain-of-Thought datasets to bake dialectal reasoning directly into the model weights natively.

8 Limitations

While the Hybrid Cascade mitigates the computational cost of processing an entire dataset through an LLM, the system still requires a minimum of 15GB VRAM to host the 4-bit quantized Llama-3.1-8B model natively. Consequently, inference latency remains higher than that of standalone encoder architectures, which could pose scaling challenges for real-time social media streaming applications. Furthermore, the TF-IDF retriever relies heavily on lexical overlap; while effective for seen targets, it may fail to retrieve optimal few-shot examples for highly implicit phrasing or purely zero-shot targets lacking distinct keywords.

References

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105, Online. Association for Computational Linguistics.

Ibrahim Abu Farha and Walid Magdy. 2020. From Arabic sentiment analysis to sarcasm detection: The ArSarcasm dataset. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*, pages 32–39, Marseille, France. European Language Resources Association.

Rasha Albalawi, Nuha Albadi, Hamzah Luqman, Saad Ezzini, Maram Kurdi, Asma Yamani, Ahmed Ashraf, Maged Al-Shaibani, and Nora Alturayeif. 2026a. Stanceeval-2026: The second stance detection shared task. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Rasha Albalawi, Nuha Albadi, Hamzah Luqman, Maram Kurdi, Saad Ezzini, Asma Yamani, and Ahmed Ashraf. 2026b. Mawqif-xt: An arabic benchmark dataset for cross-target stance detection. *Preprint*, arXiv:2608.09539.

Amal Alghamdi, A. Alshutayri, and Basma Alharbi. 2022. Deep bidirectional transformers for arabic dialect identification. *Proceedings of the 6th International Conference on Future Networks & Distributed Systems*.

Ali Alkhathlan, Faris Alahmadi, Faris A. Kateb, and H. Al-Khalifa. 2025. Constructing and evaluating arabicstancex: a social media dataset for arabic stance detection. *Frontiers in Artificial Intelligence*, 8.

Fatimah Almalki, Areej Alhothali, Lulwah Alharigy, and Abdulrahman Aladeem. 2026. Mentalmarbert: Domain-adaptive pre-training and two-stage fine-tuning for arabic mental health disorders detection. *Preprint*, arXiv:2606.12649.

Nouf AlShenaifi, Nourah Alangari, and Hadeel Al-Negheimish. 2024. Rasid at StanceEval: Fine-tuning MARBERT for Arabic stance detection. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 828–831, Bangkok, Thailand. Association for Computational Linguistics.

Loay Hatem, Ahmed Omar, Abdelmgeid A. Ali, and Heba Mamdouh Farghaly. 2025. Tackling toxicity in arabic social media through advanced detection techniques. *Scientific Reports*, 15.

Rehenuma Ilman, Mehreen Rahman, and Samia Rahman. 2025. Cuet_zenith at llms4ol 2025 task c: Hybrid embedding-llm architectures for taxonomy discovery. *Open Conference Proceedings*.

Gautier Izacard, Patrick Lewis, M. Lomeli, Lucas Hosseini, F. Petroni, Timo Schick, Jane A. Yu, Armand Joulin, Sebastian Riedel, and Edouard Grave. 2022. Few-shot learning with retrieval augmented language models. *J. Mach. Learn. Res.*, 24:251:1–251:43.

Omar Mansour, Eman Aboelela, Remon Talaat, and Mahmoud Bustami. 2025. Transformer-based ensemble model for dialectal arabic sentiment classification. *PeerJ Computer Science*, 11.

Xiaofei Sun, Xiaoya Li, Jiwei Li, Fei Wu, Shangwei Guo, Tianwei Zhang, and Guoyin Wang. 2023. Text classification via large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 8990–9005, Singapore. Association for Computational Linguistics.

Zhiqiang Wang, Yiran Pang, Yanbin Lin, and Xingquan Zhu. 2024. Adaptable and reliable text classification using large language models. *2024 IEEE International Conference on Data Mining Workshops (ICDMW)*, pages 67–74.

A Generative Fallback System Prompt

To ensure the Llama-3.1-8B-Instruct model adhered to the classification constraints and generated easily parsable outputs, we utilized the following strict system prompt during the Stage 2 fallback mechanism:

```
You are an expert in Arabic dialectal sentiment and stance detection. Your task is to classify the stance of the provided tweet towards the given target as exactly one of: 'Favor', 'Against', or 'None'.

Here are highly relevant examples from the training data:

[DYNAMIC TF-IDF EXAMPLES INSERTED HERE]

Respond ONLY with a valid JSON object containing the key "Stance". Do not provide any conversational text.
```

B Additional Error Analysis Examples

To provide further insight into the limitations of the generative fallback stage, we present additional failure cases from the validation set below. These highlight how generative models can struggle with causal reasoning and implicit desires when analyzing text.

Example 1: Contextual Negation **Target:** Digital Transformation
Tweet:
نقل الاموال مكلفة وخطرة وطبع النقود بعد مكلف عشان كذا التحول للدفع الالكتروني. الكاش يضيع او ينسرق او يتلف
Translation: *Transferring money is costly and dangerous, and printing money is also costly, therefore the transition to electronic payment. Cash gets lost or stolen or damaged.*
True Label: Favor | **Predicted:** Against
Analysis: The author lists the negative aspects of physical cash (costly, dangerous, prone to theft) to justify the transition to digital payments. The LLM detected the high density of negative vocabulary and incorrectly classified the stance as Against, failing to capture the underlying causal reasoning.

Example 2: Implicit Support via Accessibility Complaints **Target:** Covid Vaccine
Tweet:
اخذت جرعه لقاح ولا باقي للاسف ماحولنا مراكز تطعيم اقرب شي بعد ١٥٠ كيلو وان اخذته انا غيري كثير من اهل الديره مايقدرون ياخذونه
Translation:
#Did_you_take_the_vaccine_dose_or_not Unfortunately, there are no vaccination centers around us, the closest is 150 km away, and even if I take it, many others from the village cannot take it.
True Label: Favor | **Predicted:** None
Analysis: The tweet expresses deep frustration over the lack of access to vaccination centers. While the surface sentiment is entirely composed of complaints and negative context (lack of resources, distance), complaining about the unavailability of the target implies a strong desire for it (Favor). The LLM failed to make this pragmatic inference, defaulting to a neutral None stance due to the absence of explicit praise for the vaccine itself.